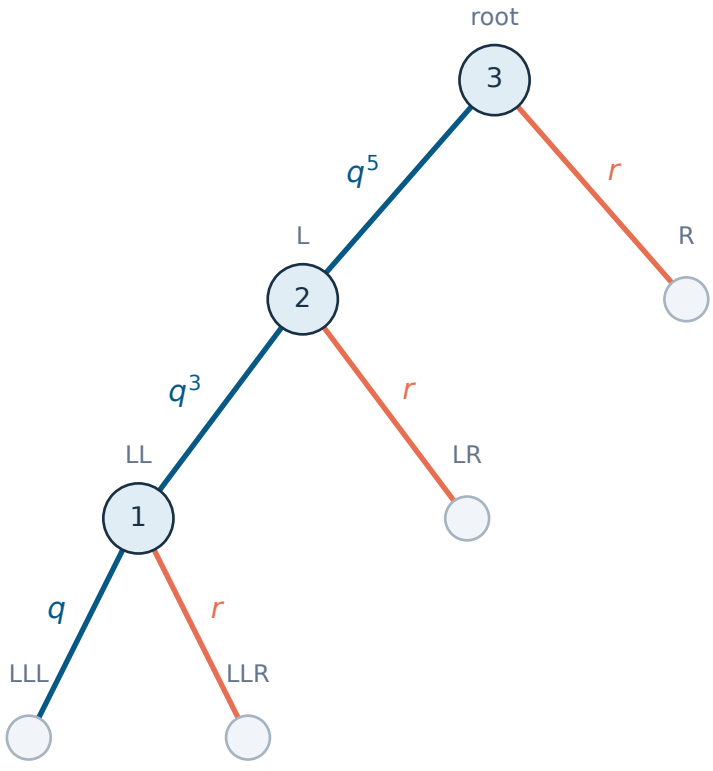
The same three nonlinear nodes have different scale addresses

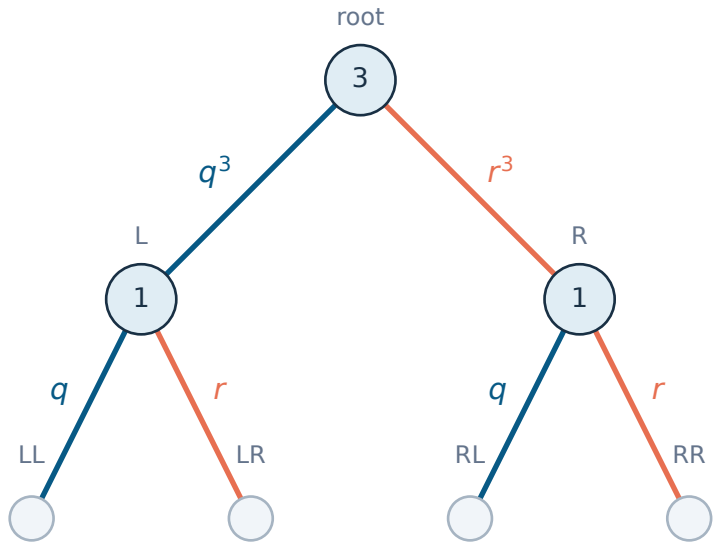
An ordered chain



Tree contribution: $48q^9r^3$

Local integer factors: $6 \times 4 \times 2$

A balanced split



Tree contribution: $80q^5r^5$

Local integer factors: $20 \times 2 \times 2$

At a node with child sizes i, j: edge weights q^{2i+1} and r^{2j+1}; integer factor $\binom{2(i+j+1)}{2i+1}$.

Node labels count internal nodes. Gray leaves carry the base term z. L and R are ordered addresses.